Hybrid III 6 y/o Lab

BME254L - Spring 2026 - Palmeri

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Table of contents

Tasks . . . 1
What to submit . . . 2
Bottom Plate . . . 4
Top Plate . . . 6
Nodding Plate . . . 8
Molded Neck . . . 10
Lower Neck Bracket . . . 13
Load Cell . . . 15
Assembly . . . 16

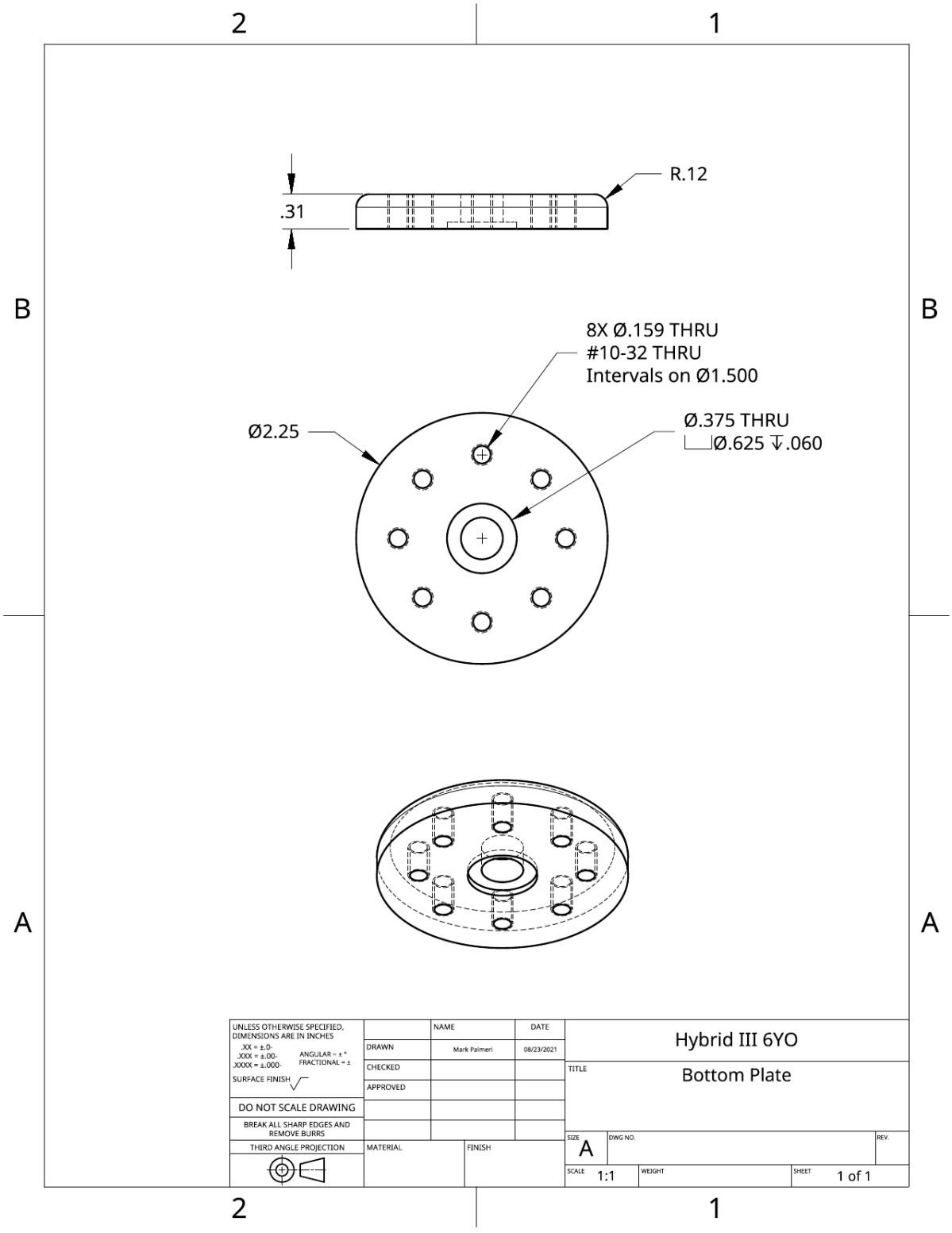
Tasks

- Create all of the parts for the Hybrid III 6yo test setup based on the attached drawings.
- You should create all of these parts and their associated drawings and assemblies as part of the same "Document" and share that document with Dr. Palmeri and the TA(s).
- Create the mechanical drawings associated with each of those parts (i.e., recreate the drawings that you are using to generate the parts) and the assembly.
- Assemble your parts as guided by the assembly diagram and as shown in the uploaded animation.
- There are online tutorials on Assemblies if you need some refreshers:
  - https://learn.onshape.com/courses/introduction-to-assembly-design
  - https://learn.onshape.com/courses/fundamentals-onshape-assemblies

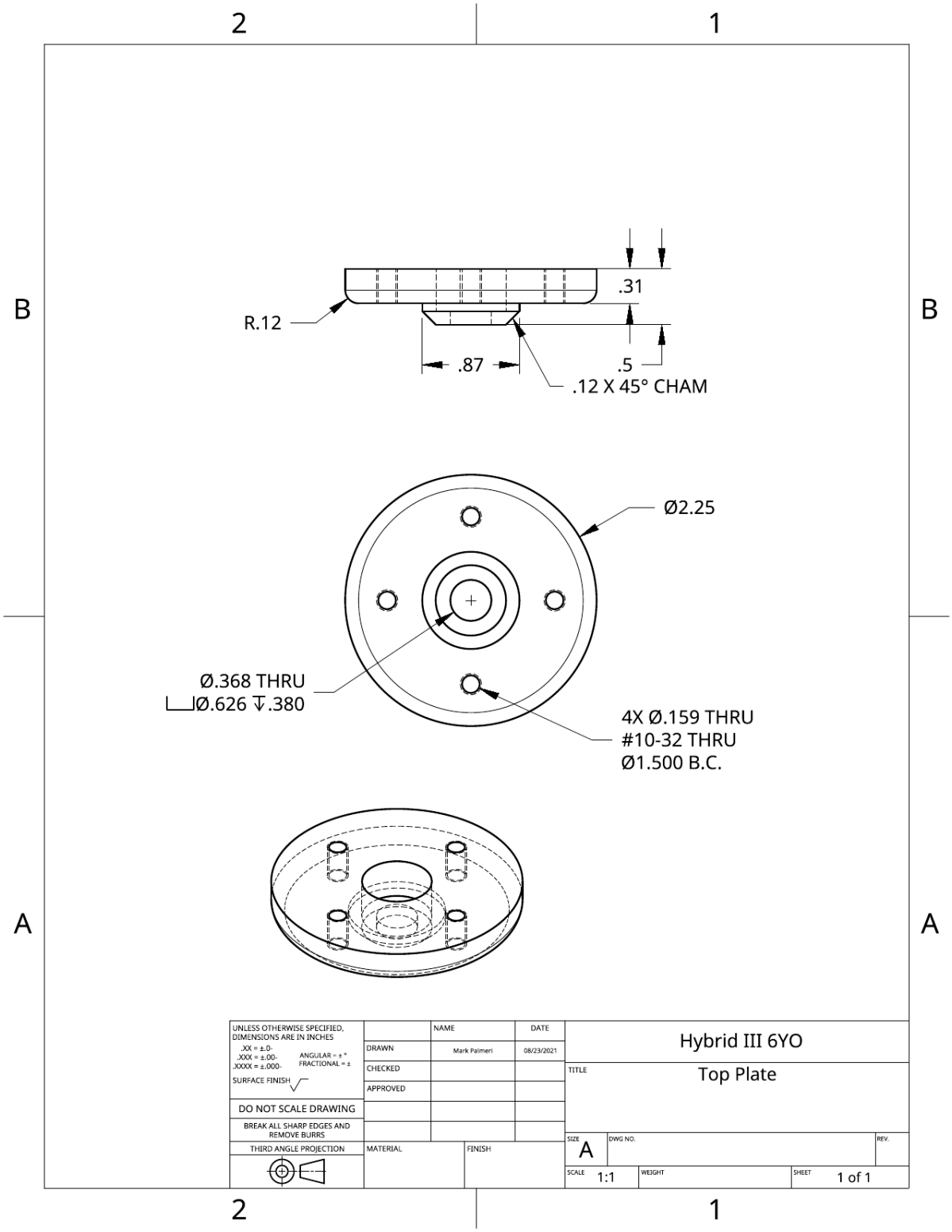
What to submit

Upload mechanical drawings of the part for all of your parts to the `Hybrid III 6 y/o` Canvas assignment. This can be done as one file per part or as a single PDF file with all of the drawings.

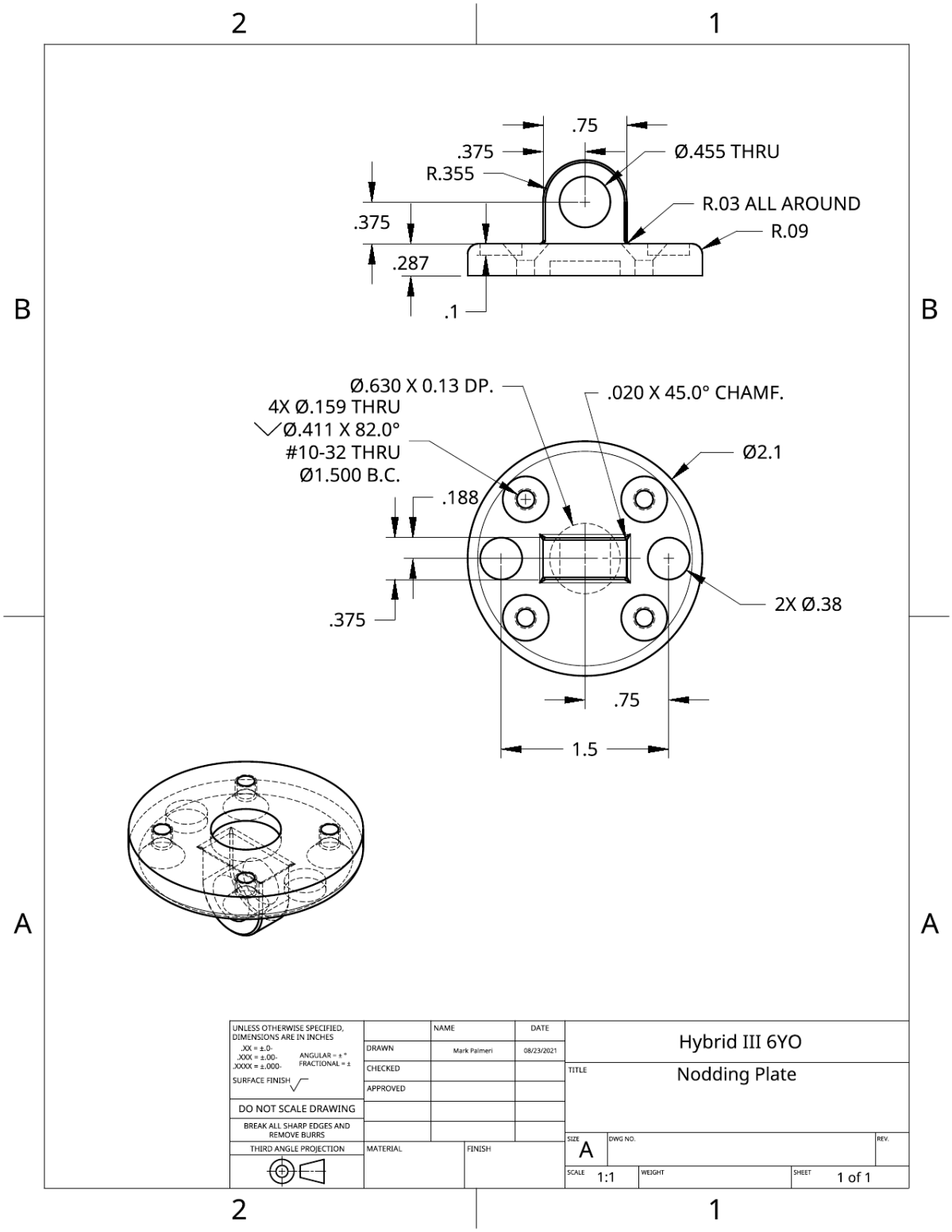
Bottom Plate



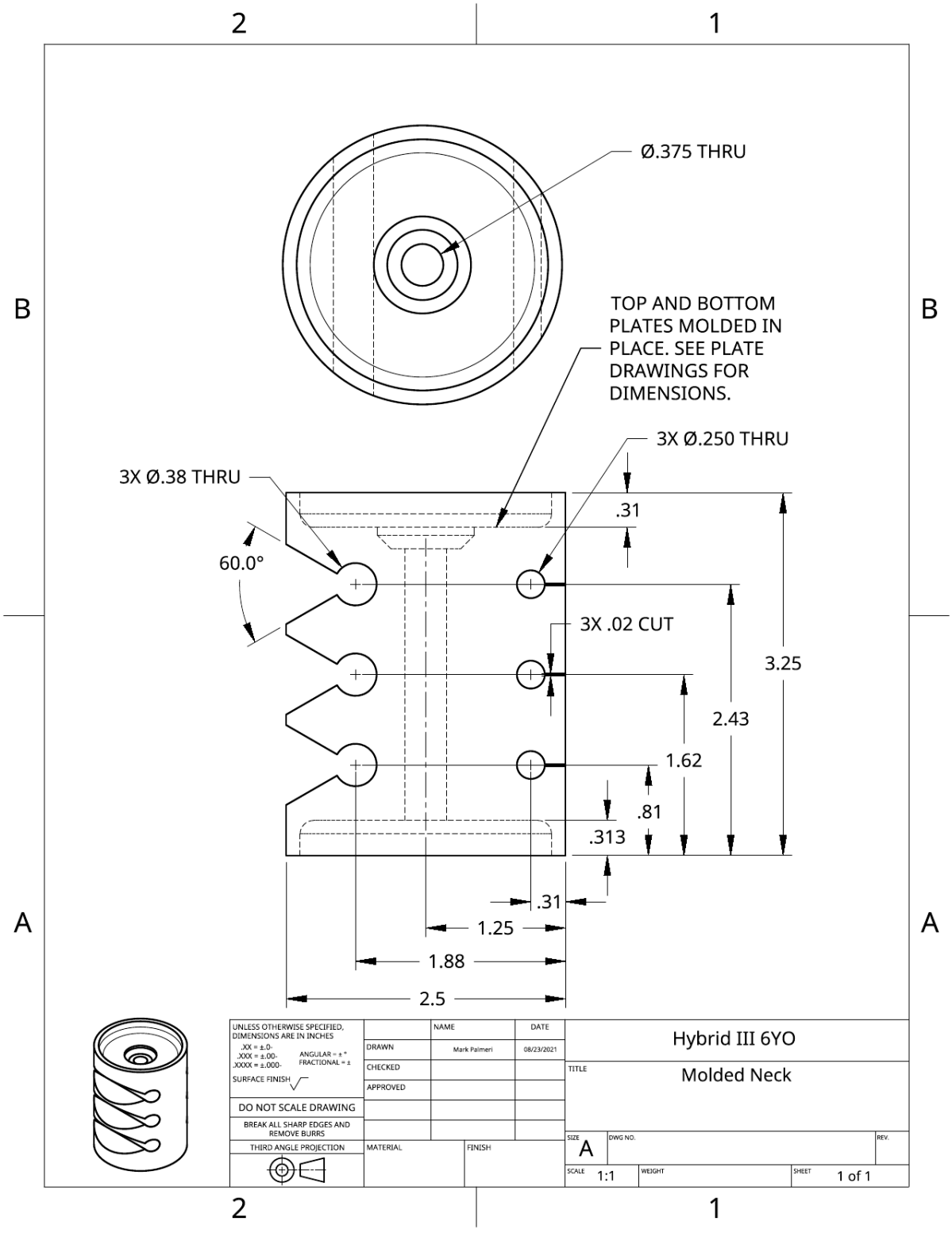
Top Plate


2
1
B
A
.31
R.12
.87
.5
.12 X 45° CHAM
Ø2.25
Ø.368 THRU
Ø.626 ↧.380
4X Ø.159 THRU
#10-32 THRU
Ø1.500 B.C.
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
Mark Palmeri
08/23/2021
CHECKED
APPROVED
MATERIAL
FINISH
Hybrid III 6YO
TITLE
Top Plate
SIZE
A
DWG NO.
REV.
SCALE
1:1
WEIGHT
SHEET
1 of 1

Nodding Plate



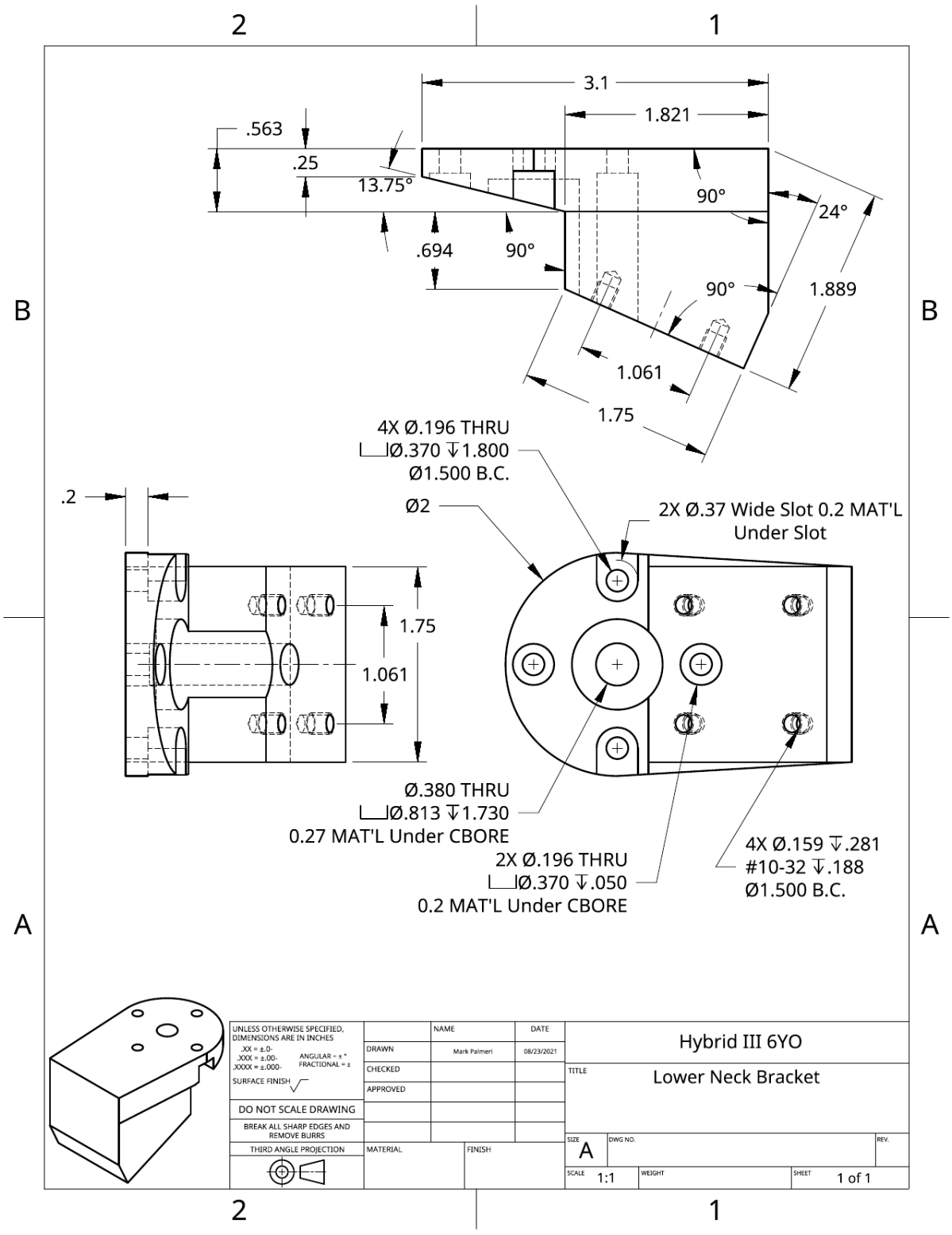
Molded Neck


2
1
B
A
Ø.375 THRU
TOP AND BOTTOM PLATES MOLDED IN PLACE. SEE PLATE DRAWINGS FOR DIMENSIONS.
3X Ø.250 THRU
3X Ø.38 THRU
.31
60.0°
3X .02 CUT
3.25
2.43
1.62
.81
.313
.31
1.25
1.88
2.5
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
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SURFACE FINISH
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SHEET
1 of 1

Tip

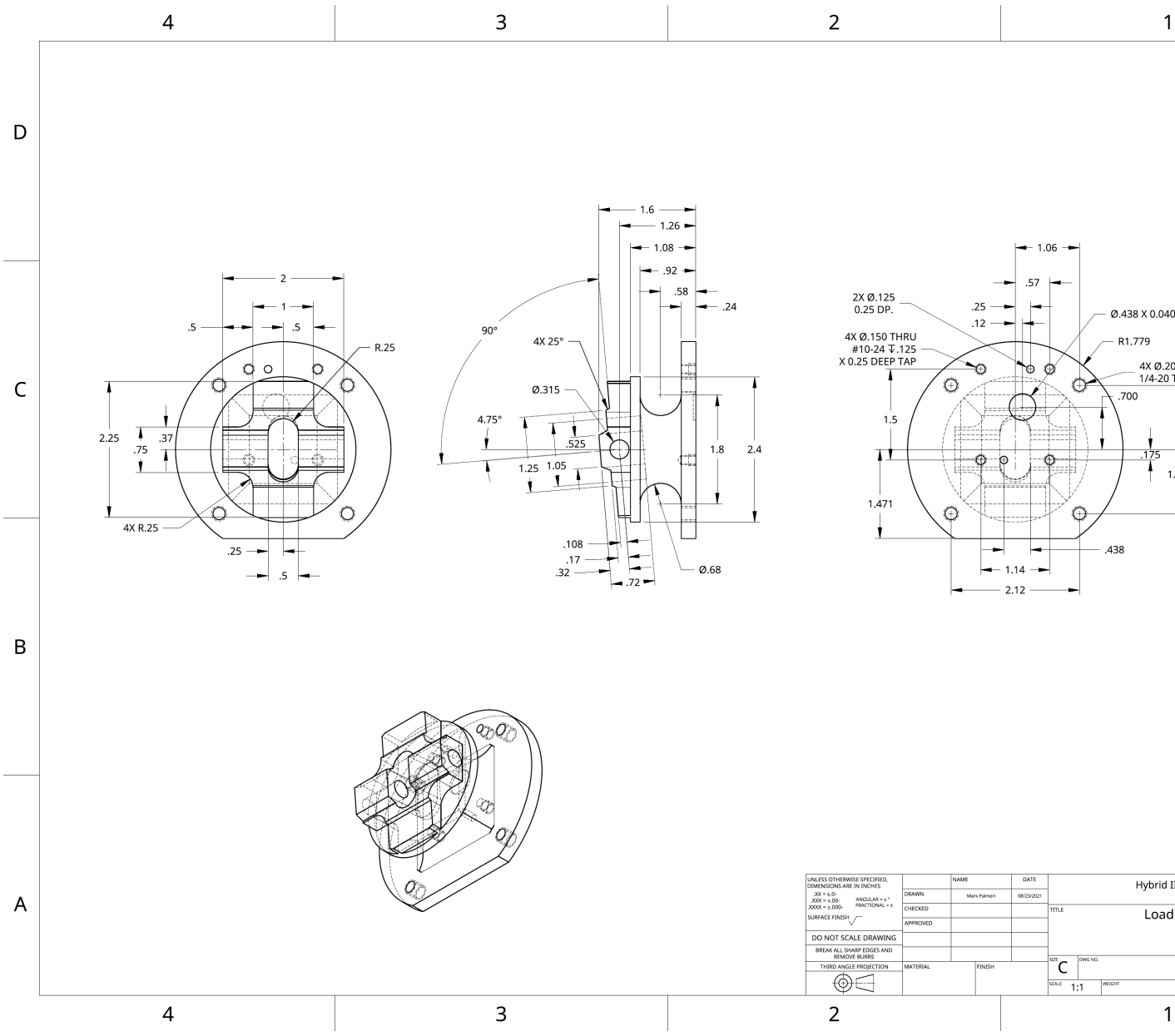
Note that the this part has features that are dependent on the top and bottom plates. To reduce the overhead of manually entering these dimensions from the other parts, you can use the `Derived` tool in Onshape to bring in the relevant faces from the other parts. Those parts do have holes in them that might create challenges, so you might want to have a version of the part without holes, a derived version with holes, and then used the non-hole part to create this derived part.
https://cad.onshape.com/help/Content/derived.htm#StepsDesktop

Lower Neck Bracket



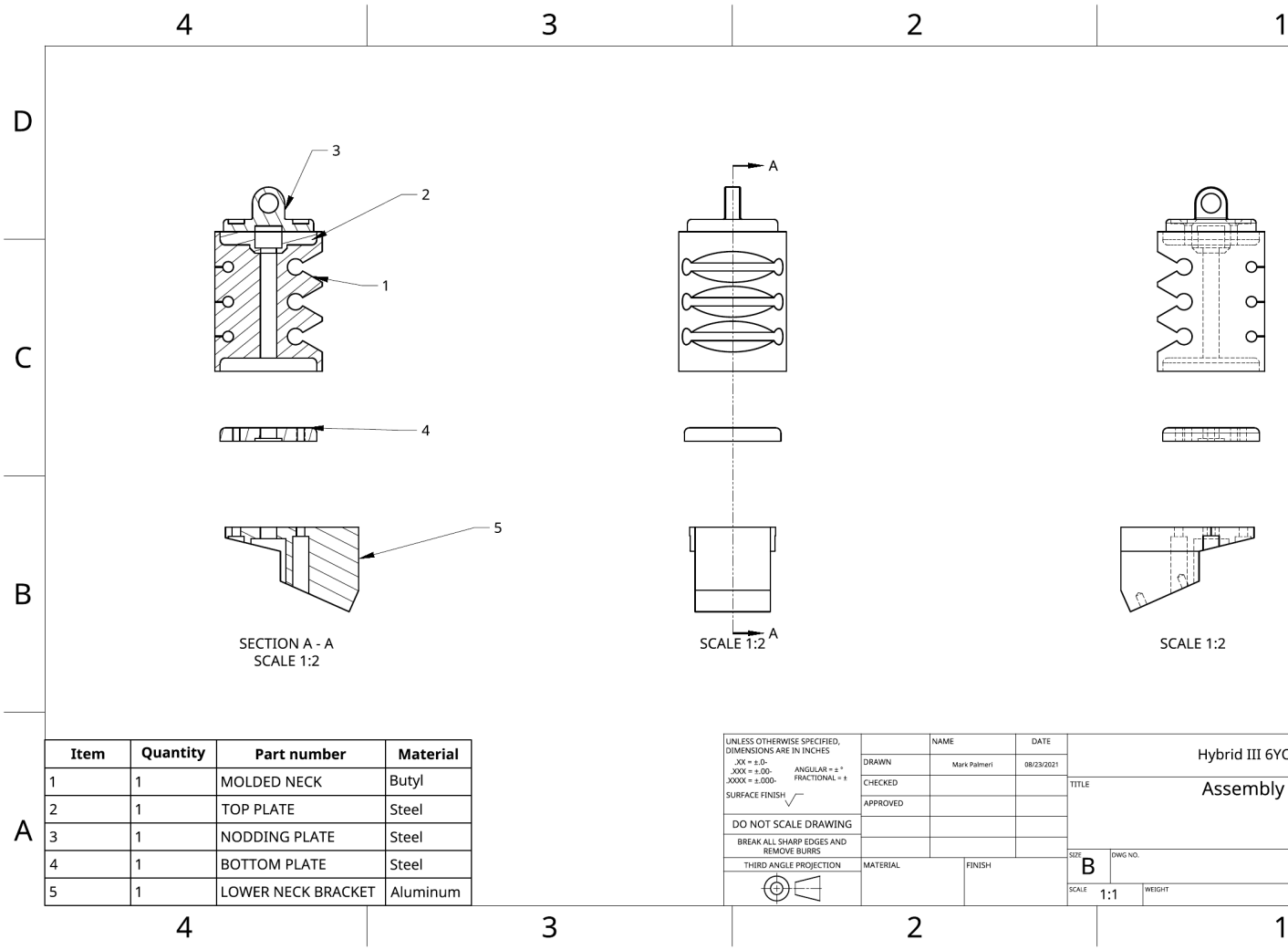
> **Tip**
>
> Note that some of the counterbore holes “start” at positions that are not on the physical part. You can determine these positions by the lengths of the counterbores and the material under the counterbores.

Load Cell



> **Tip**
>
> The `Loft` tool can be used to create the material between two part surfaces. In this case, the lofted material will need to be modified to create the curved surfaces.

Assembly



Item	Quantity	Part number	Material
1	1	MOLDED NECK	Butyl
2	1	TOP PLATE	Steel
3	1	NODDING PLATE	Steel
4	1	BOTTOM PLATE	Steel
5	1	LOWER NECK BRACKET	Aluminum

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
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ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION

	NAME	DATE
DRAWN	Mark Palmeri	08/23/2021
CHECKED		
APPROVED		

MATERIAL
FINISH

Hybrid III 6YO
TITLE
Assembly
SIZE B
DWG NO.
SCALE 1:1
WEIGHT